

Definite integrals – 8

Integrals with a parameter and passage to the limit

Homework

8 сентября 2026 года

Theoretical minimum. Continuity and differentiability of an integral in a parameter; the Leibniz rule with variable limits; term-by-term integration of a uniformly convergent series; Arzelà's bounded convergence theorem for the Riemann integral and the two standard counterexamples — a tall thin spike and a moving bump.

1. Prove that $I(t) = \int_a^b f(x, t) dx$ is uniformly continuous on $[c, d]$ whenever f is continuous on the rectangle, and give an f discontinuous at a single point for which I is discontinuous.
2. Prove that continuity of $\partial_t f$ can be weakened to: $\partial_t f$ exists everywhere and is bounded, plus integrability in x for each t . Then find where the naive proof by the mean value theorem uses exactly this.
3. (IMC) Let $f(x) = 2x(1-x)$ and let f_n be the n -th iterate of f . Compute $\int_0^1 f_n(x) dx$ for every n and find its limit as $n \rightarrow \infty$.
4. Construct integrable f_n on $[0, 1]$, uniformly bounded, converging pointwise to a function which is not Riemann integrable. Explain why this does not contradict Arzelà's theorem.
5. Prove that if $f_n \rightarrow f$ pointwise on $[a, b]$, all f_n are monotone and f is continuous, then the convergence is uniform and the integrals converge. Show that continuity of the limit cannot be dropped.
6. Let f be continuous on $[0, 1]$. Prove that $t \mapsto \int_0^1 f(x) \cos(tx) dx$ is smooth and compute all its derivatives; then prove that it tends to zero as $t \rightarrow \infty$.

7. Prove that $\int_0^1 \frac{x^b - x^a}{\ln x} dx = \ln \frac{b+1}{a+1}$ for $b > a > -1$ by differentiating in a parameter, and justify every step of the differentiation, including the behaviour of the integrand near $x = 1$ and $x = 0$.
8. Let f be continuous on $[0, 1] \times [0, 1]$. Prove that $\int_0^1 \int_0^1 f(x, y) dx dy$ may be computed in either order, using only uniform continuity and Darboux sums.
9. Give a sequence of continuous functions on $[0, 1]$ converging pointwise to 0 for which $\int_0^1 f_n \rightarrow 1$, one for which $\int_0^1 f_n$ diverges, and one for which $\int_0^1 f_n$ oscillates between two values. Which hypothesis of Theorem 3 fails in each case?
10. Let g be continuous on $[0, 1]$ and let $I(t) = \int_0^1 g(x)^t dx$ for $g > 0$. Prove that $\ln I(t)$ is convex in t , that $I(t)^{1/t} \rightarrow \max g$ as $t \rightarrow +\infty$, and identify $\lim_{t \rightarrow 0^+} I(t)^{1/t}$.